

Executive Summary: Luminous is an enterprise campus operating system uniting **high-stakes safety operations** (2-tier SOS, real-time CAD vector map, SLA timers, AI incident triage) with **complete academic ERP governance** (21 modules spanning courses, attendance, exams, hostels, placements, and wellbeing).

1 Core Technical Novelties & Architecture Innovations

- 1. Dual-Fidelity Spatial Engine & Timetable Fallback Localization**

When indoor GPS degrades inside multi-story concrete buildings (>50m-100m error), the system activates a spatial-temporal fallback: queries the student's live timetable slot (timetable_slots) and cross-references PostGIS building polygons (campus_buildings) to pinpoint the exact classroom and floor (location_source: 'timetable_fallback'), guaranteeing zero emergency blind spots.
- 2. Dynamic Multi-Tiered SOS & Autonomous SLA Escalation**

Level 1 (Campus Quick Response): Proximity dispatch to nearest guard via PostGIS ST_Distance. **Level 2 (Police & Squad Broadcast):** High-priority emergency broadcast. **Autonomous 300s SLA:** If unacknowledged within 5 minutes, autonomously escalates to Level 2. Hold-to-activate ring prevents false triggers.
- 3. Multi-Agent AI Mental Health Triage & "Warm Handoff" Protocol**

Zero-wait sentiment evaluation categorizes urgency (LOW to CRITICAL/self-harm). When transferring a student to a human therapist, AI generates a structured 3-bullet clinical summary giving immediate context while protecting anonymity and preventing traumatic repetition. Load balances therapist session queues.
- 4. Interactive Architectural Blueprint & Campus Vector Map**

Custom vector blueprint engine (MAP_VECTOR_PATH) rendering all 13 core facilities with real-time guard patrol telemetry (Available, On Patrol, Responding), dynamic pulsing radar beacons, and sector-by-sector risk intelligence heatmaps.
- 5. Confidential AI Grievance & Anti-Ragging Intelligence**

Anonymous reporting pipeline with real-time NLP classification across 8 grievance domains, severity tagging, and cryptographic isolation ensuring reporter identities cannot be traced by reviewers.
- 6. Full Convergence of Academic ERP & Campus Safety**

Unifies geofenced classroom attendance, emergency contact graphs & trusted peer circles, and cross-departmental scheduling grids (CSE, AI-DS, ECE, MECH, CIVIL) into a unified data plane with zero silos.
- 7. Zero-Trust Row-Level Security (RLS) & Multi-Persona RBAC**

Database-enforced RLS across specialized personas (Student, Faculty, Guard, Volunteer, Admin, Parent, Therapist). Therapists see only assigned clinical notes; Parents see only their child's safety status; Guards see only active incidents.

2 Frontend & Backend Technology Stack

FRONTEND STACK

- **Framework:** Next.js 16.3.1 (App Router, Turbopack, SSR/SSG)
- **UI Engine:** React 19.2.8 & TypeScript 5 (strict: true)
- **Styling:** Tailwind CSS v4.3.3 (CSS-first design tokens)
- **Component Layer:** Native React 19 primitives (zero Radix bloat)
- **Visualizations:** Recharts 3.10 + Custom SVG CAD Map
- **Icons & State:** Lucide React (800+ icons) + 4 Domain Contexts

BACKEND & INFRASTRUCTURE

- **API Routing:** Next.js Route Handlers + Edge Proxy (proxy.ts)
- **Database & Auth:** Supabase (PostgreSQL with RLS & JWTs)
- **Session Sync:** @supabase/ssr (Cookie session hydration)
- **AI Engine:** Google Gemini 2.0/3.7 Flash + Heuristic Fallback
- **Validation:** Zod v4.4 schemas on all routes & AI outputs
- **Security:** Sliding-window rate limiters, CSRF, crypto IDs

3 Stack Justification: Why These vs. Alternatives

Technology	Alternatives Rejected	Core Rationale & Advantage
Next.js 16	Vite, CRA, Remix, Nuxt	Unifies SSR, Edge Proxy, Route Handlers, and React 19 in one project. CRA is deprecated; Vite lacks native Edge proxy middleware.
Supabase (PostgreSQL)	Firebase (Firestore), AWS Amplify	Relational schema is mandatory for ERP joins (Student - Course - Attendance - Incident). Firebase NoSQL cannot handle multi-table academic queries.
Gemini Flash	OpenAI GPT-4o, Claude 3.5	Sub-200ms emergency triage latency at ~1/20th token cost. Paired with deterministic rule fallback for zero-downtime safety compliance.
Tailwind CSS v4	Bootstrap 5, MUI, Emotion	Zero runtime CSS overhead (~10KB purged bundle). Unlocks complete custom branding (Luminous Gold & Frosted Silver) without framework overrides.
Native Primitives	Radix UI, Ant Design, Chakra	Custom React 19 primitives save ~200KB bundle weight while maintaining full accessibility and eliminating external library vulnerabilities.

4 Security & Emergency Safety Architecture

SEVEN-LAYER DEFENSE-IN-DEPTH PIPELINE

- **Layer 1 (Edge Headers):** Strict CSP, HSTS (2yr), X-Frame SAMEORIGIN.
- **Layer 2 (CSRF Guard):** Rejects cross-origin fetch (sec-fetch-site).
- **Layer 3 (Rate Limiter):** SOS (10/min), Alerts (5/min), Copilot (25/min).
- **Layer 4 (Auth Guard):** Supabase Bearer JWT & cookie session hydration.
- **Layer 5 (RBAC Gateway):** 9 roles; fail-secure deny-by-default routing.
- **Layer 6 (Zod Schemas):** Strict typing & URL regex sanitization.
- **Layer 7 (Crypto IDs):** crypto.randomUUID() non-enumerable IDs.

EMERGENCY RESPONSE WORKFLOWS

- **Two-Tier SOS:** Level 1 Campus (5-min SLA) -> Level 2 Police.
- **Emergency Broadcasts:** Campus, Building, Hostel, Department scopes.
- **Sub-200ms AI Triage:** Severity classification with fallback.
- **Command Center:** Live CAD map with radar beacons & guard units.
- **Zero-Trust Copilot:** FERPA-compliant authorized query execution.

5 ERP Platform Architecture & Institutional Benefits

Luminous provides **21 integrated modules** across Academic Governance and Campus Operations:

ACADEMIC GOVERNANCE (7 MODULES) • Departments: Budget allocations, lab counts, HOD governance. • Courses & Syllabi: Credits, semesters, instructor allocations. • Faculty Management: Profiles, cabins, research, teaching hub. • Classroom Attendance: Batch marking, defaulter alerts (<75%). • Examinations: Mid-Term/Final scheduling, SGPA analytics. • Master Timetable: Grid/Agenda views with room clash prevention.	STUDENT & CAMPUS OPERATIONS (7 MODULES) • Student Directory & Portal: Roster, CGPA, daily schedule. • Parent Cockpit: Ward attendance, grades, curfew visibility. • Hostel & Curfew: Room allocations, maintenance, night logs. • Placement Services: Corporate drives, auto-CGPA eligibility. • AI Grievance Redressal: 8-domain auto-ticket routing. • Student Wellbeing: Confidential counseling & crisis lines.
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The ERP Cross-Module Advantage (Zero Data Silos)

- **Exams to Placements:** Published CGPA and backlogs in /exams automatically lock/unlock eligibility for company drives in /placement.
- **Attendance to Parent Portal:** Faculty submissions in /attendance immediately reflect in /parent for attendance defaulter warnings.
- **Hostel to Security:** Curfew turnstile anomalies in /hostel immediately trigger patrol alerts in /security.
- **Safety to Interactive Map:** Reports filed in /incidents project live hazard beacons onto the vector CAD blueprint in /campus-map.

6 Future Expansion Roadmap

NEAR-TERM (PHASE 1) • Web Push / PWA: Emergency push alerts even when browser is closed. • Dark Mode UI: Night-shift theme for security desk operations. • PDF/CSV Export: Attendance sheets, grade cards, audit logs. • SMS/Email Fallback: Twilio SMS for critical campus emergencies.	MEDIUM-TERM (PHASE 2) • Mobile Native Apps: React Native / Flutter apps with offline SOS. • CCTV Camera Stream: Real-time feeds in Command Center with AI anomaly alerts. • Tuition & Fee Gateway: Payment reconciliation and scholarship tracking. • Fleet GPS: Live campus shuttle tracking and QR transport passes.	LONG-TERM VISION (PHASE 3) • Smart Building IoT: Smoke/gas detectors feeding directly into Safety Context. • Multi-Campus Federation: Centralized governance across university branches. • AI Vision Surveillance: Automated perimeter breach & crowd analytics. • Voice-Activated SOS: Hands-free voice emergency trigger ("Hey Luminous").
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